Name: Ihit Rajesh Acharya
Reg: 240958198
Roll: 48
Fintech-A

Research Paper

Reinforcement Learning from Delayed Observations via World Models

This paper focuses on how to do RL when an agent's observations have delay, and tackles the harder and less studied case where the underlying environment is only partially observable MDP rather than fully observable MDP. The idea used is world models, which are learned internal representations that compress observations into a latent state and learn the dynamics of that state to cope with the delay.

The paper suggests a Delayed POMDP as a POMDP paired with a distribution over delay lengths, so that at time t the agent receives the observation and reward that were actually generated at time ‘t-d’.It shows that DMDPs are a special case of DPOMDPs. If a world model is aligned with a POMDP then the "delayed world model" is same as the delayed POMDP. This justifies training the world model on ordinary and undelayed transition data collected during delayed interaction, and only handling delay explicitly in the policy learning stage, since the world model already remembers history and embeds the effect of delay into its latent state.

The paper also gives a small MDP example showing that delay can degrade optimal value badly depending on the environment's randomness, showing that the ratio between delayed and undelayed optimal values depends on the discount factor and a probability parameter, and that this gap is more severe at intermediate levels of environment randomness.

Methods proposed. Building on Dreamer style world models (encoder to RSSM to decoder architecture, alternating world-model learning and imagination-based actor-critic learning), the author also compare several strategies for adapting a world model to delayed settings:

- Extended actor: the policy is conditioned on an extended state consisting of the last available latent state plus the sequence of actions taken since then, while the critic still evaluates the current undelayed latent state, so the imagination length is lengthened by the delay. A memoryless actor variant drops the explicit action history,suggesting that the network can implicitly encode it since no new information arrives after the last observation.

- Latent actor: Rather than changing the policy architecture, the current latent state is estimated by simulating the world models learned dynamics for the next d steps starting from the last observed latent state, and this state is fed to an zunmodified undelayed policy.

- Agnostic baseline: a world model trained entirely without awareness of delay is naively applied at test time under delay, it serves as a naive baseline to show what happens without any delay specific adaptation.

The methods are implemented on top of DreamerV3 and evaluated across both continuous-control tasks (MuJoCo / Gymnasium tasks) and, visual (image-observation) tasks from the DeepMind Control Suite which they present as the first delay aware RL evaluation using visual observations, since image inputs

are partially observable in a way vector states are not. Baselines for comparison include prior delay methods designed for fully observable delayed MDPs (an extended approach using normalizing flows for belief representation, and an approach that resamples an extended state with an off policy actor critic model). The paper also runs an test exploring how the different delay handling strategies degrade as the underlying environments degree of partial observability is varied, to see which methods are robust to compounded delay and partial observability.

Key findings are naively applying a delay unaware world model at test time works reasonably well in some low-stochastic envs but performance drops fast as delay grows, because the training and testing observation distributions diverge more under longer delay. Feeding the policy raw latent state predictions works well for short delays but degrades for long delays because one step prediction errors from the model compound over many forward imagination steps. The Extended actor, despite adding architectural complexity was found to be the most robust variant to both increasing delay length and increasing partial observability, because it uses the actual known action sequence rather than relying purely on the world models own error rollout, hence avoiding the compounding error problem of the Latent approach and the loss of memory in the Memoryless approach. The paper concludes that Dreamer style world models can be adapted to delayed and partially observable envs without needing domain specific hyperparameter tuning, offering better or comparable performance to prior delay specific methods that were designed only for fully observable delayed MDPs, and that the choice of adaptation strategy involves a tradeoff between architectural simplicity and robustness to longer delays.

Deep Reinforcement Learning for Optimal Trading with Partial Information

This paper addresses optimal trading of a signal that follows a mean reverting process(Ornstein Uhlenbeck process) whose parameters (long run mean, speed of mean reversion, volatility) are latent and also switch between regimes according to independent Markov chains. The central question is how best to give a RL trading agent access to information about these hidden regime switching parameters, and how the quality of that information affects the trading policys performance.

The signal dynamics are modeled as a continuous-time mean-reverting SDE, which is discretized for the RL model usage. The trading agents action at each time step is the new inventory level to hold and the reward is the mark to market change in position minus a transaction cost proportional to trade. The paper frames this as a partial information stochastic problem, the agent must simultaneously infer the hidden regime while deciding how to trade. Complexity of env is increased across three stages being first only the long run mean parameter follows a MDP, then both the mean and the speed of mean-reversion follow independent MDP, and finally all three parameters (mean, speed, volatility) are regime switching independent MDP.

The paper proposes three DDPG based algorithms, all using a GRU network to encode the time dependency of observed signal:

- hid-DDPG: the GRUs hidden state is fed directly, alongside current signal value and inventory, into the DDPG actor critic and the GRU is trained jointly with the trading policy, so filtering and trading are learned together in one pass.

- prob-DDPG: first a GRU based classifier is trained offline (using knowledge of the true latent regime

during training) to output probabilities of being in each regime given the signal history, then a separate DDPG agent takes these probability estimates, plus the current signal and inventory, as its feature/state input for learning the trading policy. This decouples the filtering step from the trading optimization.

- reg-DDPG: similar decoupling, but instead of learning regime probabilities, a GRU is first trained to forecast the next value of the signal, and this forecast is used as the extra feature fed to the DDPG trading agent. This serves as a natural benchmark for whether raw prediction is as useful as probabilistic regime information.

The three algorithms are first tested on simulated data across the three increasingly complex MDP comparing cumulative reward distributions over many episodes. The paper also performs a qualitative analysis, visualizing the buy/sell action chosen as a function of inventory and signal level to assess whether the learned policys actions align with the mean reversion. Beyond simulation, the paper applies the best performing approaches to two co-integrated NASDAQ listed equities and regimes are estimated using a Hamilton style MDP model, the RL trading policies are then trained on this real signal and tested, compared against a classic Z-score trading strategy benchmark.

Across both the synthetic experiments and the real pair trading application, providing the agent with probability information about the latent regime consistently produced the best trading policies because it lets the agent condition its buy/sell decisions on both the current inventory level and the most likely regime. Simply giving the agent regime trained estimated MDPs provided intermediate performance and the network could still exploit encoded information, but the resulting policy was harder to interpret due to complexityy of encoded learned regime MDP. Providing only point forecast of the next signal value (reg-DDPG) gave least benefit and performance degraded as the parameter dynamics became more complex, since single point prediction failed to convey the regime information relevant to a mean reversion trading decision, especially when there is more uncertainty.

DSA: Evidence-Aware LLM-Agent Orchestration for Multi-Market Stock Research

This paper presents DSA, a systems/architecture paper describing how to orchestrate LLM agents to conduct stock research across multiple international markets with an focus on how evidence, unequal model capabilities, and generated opinions should flow through a pipeline in a controlled and inspectable way rather than treating stock research as a single question call to an LLM.

They proposed a five stage evidence to report workflow.:

1) Evidence acquisition

2) Structured context construction: Technical features are computed and evidence blocks are built, with explicit degradation states so that absence of data is not co-inflated with a neutral or negative signal.

3) Model and capability routing

4) Optional role reasoning and controlled synthesis: Role based agents are admitted via configured prompts and passed through role specific parsers, while Strategy models pass through an additional post-generation signal validity check before their opinions are allowed to enter the evidence and the decision agent.

5) Report and inspection record

The framework formalizes the agentic synthesis mathematically,and computes an explicit disagreement summary between agents, feeds that disagreement summary plus the evidence and opinions into a decision synthesis step producing an initial decision, and then again applies a separate post synthesis Riskguard step that preserve or downgrade the decision's aggressiveness but never promotes it, this encodes a conservative risk control model. The paper stressed about strategy skills being treated as trusted external extension inputs.

The contribution is a frozen, claim mapped software test eval, not an eval of analytical quality or returns. At a specific revision snapshot, a large manifest of portable contract tests was run and a subset of these tests was mapped onto six contract families that directly correspond to the agent architecture. The paper repeated stating what this evaluation does and does not, it showed that the described control mechanisms are implemented and exercised by tests but it does not prove that reports are factually correct, that the agentic profile is better than the simpler default report profile, that one underlying LLM is better than another for this task, or that any of this translates into better investment outcomes, those comparisons are explicitly named as open future work rather than claims made by this paper.

The papers contributed to architectural model orchestrating and evaluative methodology for keeping evidence quality, tool capability, and generated opinion all observable and controlled inside a multi market, LLM agent pipeline, together with a structured way of reporting what a software test suite can and cannot be taken to evaluate such a system.

Non-obvious Manipulability with Groups in Shapley–Scarf Housing Markets

This is a market modelling paper set in the Shapley scarf housing market, where each agent owns one indivisible house and agents may trade with each other according to their preferences over all houses, with no money involved. The paper aims to explore the mechanisms behind the incentive requirement of strategy, when relaxed to a weaker non-obvious manipulability condition, and to evaluate how much freedom that relaxation buys the mechanism designer.

The result being built upon top trading cycles (TTC) mechanism satisfies individual rationality (IR), Pareto efficiency (PE: No allocation could make someone better without making someone else worse), and strategy proofness (SP: truthful reporting is always a dominant strategy. The paper premise is that full strategy proofness may be more than actually required, since real agents may not be able to reason through every possible profile of other agents reports. This motivates the usage of non obvious manipulability(NOM), a mechanism design, under which a manipulation only counts as a problem if it is obvious relative to truthful reporting.

Standard NOM is too weak in this domain, paper proves that in Shapley markets, any mechanism that is merely individually rational and Pareto efficient is automatically NOM, meaning the weaker NOM condition imposes essentially no additional changes, IR and PE are already assumed, because the presence of an initial house pins down each agents worst-case and PE already makes the agents best-case outcome reachable by appropriate cooperation of the rest of the market, so no misreport/manipulation can improve either extreme. This negative structural result motivates the rest of the paper, since bare NOM is not restrictive enough to be an incentive condition here, a strengthened notion is needed.

New concept: NOM with groups. The papers main technical innovation is to partition agents into given random groups, and to define NOM with groups requiring that no agent can obviously their best or worst case outcome by misreporting when they know exactly what thier own group members will report but faces uncertainty about the reports of everyone outside the group. This condition interpolates smoothly between existing notions, if there is a single group containing everyone, it collapses to ordinary strategy-proofness. The pape distinguishes this from group strategy-proofness (which is about ruling out coordinated joint deviations by a group), stressing that in their model the deviation is by a single individual, the groups role is purely informational.

Group rationality (GR), the paper introduces GR, requiring that every agent obtain an outcome at least as good as what she would receive under the TTC allocation if her group traded only among itself (no group is worse off for having joined the larger market). This is motivated by a real-world applications like kidney exchange, where a group can represent a hospital or country, and has documented that hospitals may strategically withhold easy to match patient donor pairs from a larger exchange pool unless they are guaranteed to not be made worse off by participating.

The paper defines super-groups which are unions of the base groups and defines it as a staged mechanism that repeatedly applies ordinary TTC to progressively larger group, continuing until the entire market has traded. This produces a family of mechanisms that differ from and generalize. The paper explicitly illustrates that these staged mechanisms are not strategy-proof, showing that intermediate information can create profitable individual misreport opportunities that would normally not exist under one shot global TTC.

The paper proves that every mechanism in the TTC with super groups class satisfies GR, PE, and NOM with groups simultaneously and proves that all TTC-with-super-groups mechanisms are equivalent to one another. The paper's central characterization result then shows the converse in spirit at the envelope level any mechanism satisfying GR, PE, and NOM with groups must induce exactly the same best-case/worst-case as every TTC with super groups mechanism.These conclusions do not pin down a single unique mechanism they do pin down the extreme outcomes available to every agent, leaving room only for variation in intermediate, non-extreme scenarios.

The paper results are as folllows: showing that relaxing SP to group based, best/worst-case notion opens up a larger space than sticking to full SP while still preserving strong PG and pinning the boundary results that each agent could expect.

Business Arena: Benchmarking LLM Agents in a Realistic Marketplace

This paper introduces Business areana, a simulation environment in which an LLM agent operates a cross border B2B shop over a long time horizon, to test whether LLM agents can function as genuine end to end economic operators rather than merely completing isolated business workflows.

Four properties that make running a business fundamentally different from typical agent benchmarks: are that evidence is noisy, feedback is delayed and hard to attribute, the environment changes continuously and independently of the agent's own actions and operational obligations persist. The authors say that these properties make direct testing in a live real business unsuitable as a first evaluation motivating a controlled but realistic simulation instead.

The arena grounds its data in real Alibaba.com sourcing listings and calibrates demand cycles, tariffs, and market conditions from trusted authoritive external sources rather than using purely synthetic numbers. An episode begins with a pre opening setup phase followed by many simulated operating days, each day the agent may inspect the market, adjust listings, replenish inventory, allocate advertising spend, manage liquidity, and respond to buyers/suppliers, after which it advances the simulated day, at which point the rest of the market evolves autonomously and independently of the agent. The paper exposed more than 60 tools spanning market research, sourcing, inventory, pricing/listing, advertising, customer service, compliance, and finance, accessible both as fine grained typed calls for interleaved reasoning and action and as a persistent workspace supporting model analysis scripts and automated operating routines. The agent runs as an unprivileged user with no access to the simulator own code, database, or hidden state, interacting with the market only through its public interface. The arena also models autonomous NPC competitor sellers with a range of distinct scripted pricing and behavior structures, calibrated with realistic scale and capital tiers, so the agent competes in a heterogeneous marketplace rather than against a single fixed opponent policy.

Financial realism is enforced through a ledger backed accounting layer so that final scoring reflects true realized business value rather than exploitable timing quirks like unsold inventory being counted at face value,etc.

Drawing on management decision theory, the paper organizes the skills into four capability areas being decision-making under uncertainty, strategic planning under constraints,insight-to-action alignment and cooperation and competition.

Because a single terminal profit number conflates many different underlying decisions and cannot by itself explain success or failure, the papers eval methodology has several layered components:

1) A library of deterministic expert designed reference strategies built using only the same publicly available agent information, used to estimate how much opportunity was actually available in a given simulated market realization, so that a model score can be judged against an achievable benchmark rather than in isolation

2) Skill level diagnostic metrics decomposing performance into submetrics for each business function which reveal distinct recognizable operating regimes that are invisible in the aggregate net worth ranking alone

3) Action level tooling traces specific realized economic gains/losses back through sequence of evidence, calculation, and action that produced them. It is intended to support fine grained credit assignment.

4) Stateful evaluation mechanism that can restore the exact business state at a chosen checkpoint, enabling controlled continuation and enabling a runtime compute technique called trace search(forking multiple continuations from a checkpoint and periodically pruning down to most promising ones).

To guard against the risk, a realistic looking benchmark could reward simulator shortcuts rather than business skills, the paper conducts mechanism ablations for several mechanisms. In each case they hold the rest of the operating policy fixed and compare an intended, evidence using policy against both a neglect and a shortcut variant checking whether the intended, competent behavior in fact outperforms both.

Because long-horizon simulated business outcomes are highly variable the paper studies the statistical reliability of its leaderboard, it reports that model identity explains a substantial part, but not all of the

variance in outcomes across runs of the same model, it also computes measure of how much repeated sampling reduces this trajectory-level noise, and performs a split resampling analysis showing that independently estimated rankings from disjoint subsets agree with each other, supporting the claim that the model ranking differences reflect model capability differences rather than a few lucky trajectories noting this reliability claim is conditional on the shared evaluation world used and that testing robustness across genuinely different world realizations is a separate open question.

The paper is states that the arena evaluates decision making abstracted behind tool calls, not an agents ability to operate real, changing external env with real side effects, not yet addressed, evaluation setting. The arena is also explicitly scoped to cross border B2B commerce, described as one slice of the much broader space of real business scenarios that could be modeled in the future.

Paper 1

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Paper 5

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